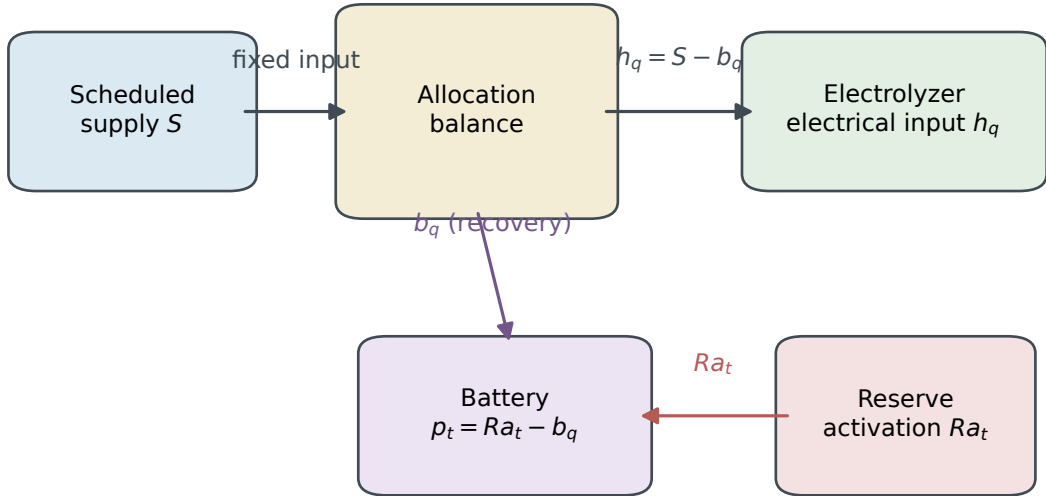


Power balance: $S + p_t = h_q + Ra_t$



Negative b_q can draw battery energy to the electrolyzer; downward activation can import additional grid electricity.